



AERODYNAMICS

The most critical part of an F1 car, because it defines the shape, the overall design, and how capable the car is of sticking close to the ground at speeds above 200 km/hr. Also, the aerodynamics is important since the direction of the airflow helps improve the speed of the car and the stability around the corners and quick bends.



KERS

This is the Kinetic Energy Release System that has now been implemented in F1 cars. Developed by Magneti Marelli of Italy, an electric motor converts kinetic energy into electrical energy, which is then stored in a separate battery. This charge comes because of the energy created by the brakes. The charge is used to power the car, for a select duration and on select sectors on every lap, adding 85bhp more power to the engine.



SAFER STRUCTURES

F1 drivers usually escape unhurt even after the most serious of accidents. This is because of the carbon composite materials used, making the car extremely safe, and lighter at the same time.

THE BRAKES

Surface temp of brakes immediately after a race- 1000 degrees C

MECHANICAL

The F1 car is an engineering marvel, and an extremely powerful one at that. The combination of engine, gearbox, tyres, brakes, suspension, steering wheel and even the fuel tank work together make the car work. The extremely powerful engines, V6s at the moment, can rev up to 15000rpm. The gearbox can change gears in milliseconds. The temperature of the brakes can go beyond 1000 degrees C. The tires can be changed in less than 5 seconds, individually. The fuel tank is made of Kevlar that prevents it from getting punched or torn open in case of a crash.



THE TYRES

- Use nitrogen instead of normal air
- Can withstand speeds over 300 km/hr without burning out